

# Aly Mahmoud

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## SUMMARY

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AI Engineering student at Ain Shams University (GPA 3.7) with strong foundations in Machine Learning, Computer Vision, and emerging LLM systems. Experienced in building end-to-end AI pipelines, including real-time computer vision systems and retrieval-augmented generation (RAG) and agent-based architectures. Delivered production-grade solutions achieving 87–99% performance across vision and NLP tasks. Eager to apply hands-on AI engineering skills to real-world, scalable systems.

## TECHNICAL SKILLS

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**Programming:** Python, C/C++, Java

**Machine Learning:** Scikit-learn, XGBoost, K-Fold Cross-Validation, SMOTE, K-Means, Feature Engineering, Model Evaluation.

**Deep Learning:** PyTorch, TensorFlow, Keras, Transfer Learning.

**Computer Vision:** OpenCV, YOLO, Object Detection, Image Classification, GANs, Multi-Object Tracking, ByteTrack, Optical Flow, Perspective Transformation

**NLP & LLMs:** NLTK, TF-IDF, Sentence Transformers, RAG, Prompt Engineering.

**Data & Visualization :** NumPy, Pandas, Matplotlib, Seaborn.

**Tools & Deployment:** Git, Streamlit, FastAPI, FAISS, LangChain, Jupyter, Google Colab

## EXPERIENCE

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### Computer Vision Trainee

Jan. 2026 – Feb. 2026

*Instant X Orange*

*Giza, Egypt*

- Built and developed 5+ deep learning models for image classification, segmentation, and object detection tasks.
- Applied transfer learning using 3+ pre-trained architectures, including YOLO, achieving faster convergence and real-time detection performance
- Explored 2+ advanced computer vision techniques, including GANs, focusing on image generation and data augmentation use cases.

### Machine Learning Intern

Aug. 2025 – Sep. 2025

*Elevo Pathways*

*Cairo, Egypt*

- Architected 4 production-level ML pipelines using Streamlit, resulting in real-time predictive tools for finance and environmental sectors
- Designed and deployed end-to-end ML pipelines (data preprocessing → feature engineering → model training → deployment) to simulate industry workflows.
- Optimized model fairness by implementing SMOTE and cross-validation, achieving a 30% increase in minority-class recall.

## PROJECTS

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### Retrieval-Augmented Generation (RAG) System | FAISS, Sentence Transformers, FastAPI

May. 2026

- Developed a RAG pipeline using embeddings and vector search for knowledge-grounded response generation
- Indexed and queried 50K+ documents using FAISS, achieving sub-second retrieval latency
- Implemented coarse-to-fine retrieval (bi-encoder + reranking), improving answer relevance by 25%
- Optimized chunking and prompt design to reduce hallucinations and improve response accuracy

### LLM Agents with Tool Execution | LangChain, OpenAI API, FAISS

Apr. 2026

- Built LLM-powered agents capable of tool calling (APIs, Python functions) to execute multi-step tasks autonomously
- Implemented a ReAct-style reasoning loop, enabling dynamic decision-making and action execution across tasks
- Integrated external tools (search, APIs, scripts), successfully handling multi-step queries
- Reduced response latency via caching and optimized prompt design

- Football Analysis System** | *YOLO, OpenCV, OpticalFlow, ByteTrack, KMeans* Mar. 2026
- Engineered an end-to-end computer vision pipeline processing 1,750+ video frames, detecting and tracking 22 players, 1 referee, and the ball with 87%+ confidence using dual YOLOv8x models specialized for player and ball detection separately
  - Reduced unique player track IDs from 800+ to 28 via a custom position-based stable ID remapper on top of ByteTrack, implemented Lucas- Kanade optical flow for camera movement compensation and OpenCV perspective transform converting pixels to meters.
  - Automated team assignment via KMeans kit-color clustering, real-world speed estimation (km/h), Team possession distribution (Team A, Team B, Neutral), and exported per-player statistics: distance covered, avg speed, ball touches and team possession breakdown to CSV.
- Movie Reviews Sentiment Analysis** | *NLTK, Pandas, NumPy* Dec. 2025
- Processed and cleaned 50,000 IMDB movie reviews, applying tokenization, stopword removal, and POS-aware lemmatization as part of a full NLP preprocessing pipeline.
  - Implemented TF-IDF vectorization from scratch using Python and NumPy, without relying on scikit-learn's vectorizers.
  - Trained and evaluated Logistic Regression and SVM models, achieving up to 90% accuracy on an 80/20 train-test split.
- ExoPlanet Classification - Nasa Space Apps Challenge** | *Flask, React, Matplotlib, SQLite* Oct. 2025
- Developed a full-stack web application (Flask backend, React + Tailwind frontend) for automated exoplanet detection using NASA's Kepler, K2, and TESS datasets.
  - Achieved 95%+ accuracy, precision, and recall across multiple machine learning models through data preprocessing, skew resolution, and hyperparameter tuning.
  - Integrated interactive dashboards, real-time prediction, and Jupyter notebook execution for end-to-end analytics and visualization in exoplanet classification.

## EDUCATION

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### Ain Shams University

*Bachelor of Computer Science, Specialized in AI, CGPA: 3.7*

Cairo, Egypt

*Aug. 2023 – Present*

### East London University

*Bachelor of Computer Science, Dual Degree*

London, England

*Aug 2023 – Present*